

Kinematic Basis of the Linkage Simulator Prototype

Implementation-derived technical note

1 Scope and conventions

The prototype is a deterministic planar kinematics demonstrator with a bounded quasi-static gravity estimate, not a validated clinical or musculoskeletal model. The serial-link abstraction follows established hand kinematic modelling practice [1, 2]. World coordinates use millimetre-like units, angles use radians internally, and positive angles are counter-clockwise. A rigid link of length L has a centre pose $(\mathbf{p}, \theta)$ and local endpoints $(-L/2, 0)$ and $(L/2, 0)$. A local point $\mathbf{q}$ maps to world space as

$$\mathbf{x} = \mathbf{p} + R(\theta)\mathbf{q}, \quad R(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}. \quad (1)$$

The data model permits arbitrary collections of links and revolute joints. The first solver strategy is deliberately narrower: one four-bar topology identified by stable component IDs.

2 Independent digit workspaces

The application instantiates independent simulation states for D2 through D5 and renders only the state selected by the active tab. Servo angle, animation direction, linkage edits, contactors, joint-limit status, and static-moment results therefore persist independently for each digit. A shared hand-size factor h applies to all four anatomical models. Neutral initialization uses rounded between-digit scales

$$(s_2, s_3, s_4, s_5) = (1.00, 1.11, 1.03, 0.83), \quad (2)$$

consistent with reported young-adult mean digit lengths of approximately 63.9, 70.7, 65.5, and 53.3 mm, respectively [3]. For base D2 segment length $L_{i,2}$, digit d is initialized with

$$L_{i,d} = h s_d L_{i,2}. \quad (3)$$

The linkage template is scaled by s_d at creation so the four tabs retain the same dimensionless mechanism geometry. Subsequent hardware edits remain local to their digit and are not automatically resized when h changes.

3 Dorsal mounting geometry

Let $\mathbf{G}$ be a point on the dorsal hand-adjacent ground plane, with unit tangent $\mathbf{t}_g$ and dorsal normal $\mathbf{n}_g$. The editable servo-ground offset h places the servo axis at

$$\mathbf{A} = \mathbf{G} + h\mathbf{n}_g. \quad (4)$$

The fixed dorsal rail has editable length L_g and servo-relative angle α_g . Its second endpoint, which grounds the proximal rocker, is

$$\mathbf{D} = \mathbf{A} + L_g(\cos \alpha_g, \sin \alpha_g). \quad (5)$$

Thus the rocker anchor is visibly structural rather than a floating ground symbol. Repositioning the servo preserves the ground-plane normal reference; editing h , L_g , or α_g deterministically rebuilds both ground pivots before the mechanism solve.

4 Analytic four-bar solve

Let $\mathbf{A}$ be the servo pivot, $\mathbf{D}$ the rocker ground pivot, and r_1, r_2, r_3 the crank length, crank-to-rocker joint distance along the coupler, and rocker length. Both ground pivots are dorsal and proximal to the D2 metacarpal; there is no distal ground pivot. Given servo angle ϕ , the crank endpoint is

$$\mathbf{B} = \mathbf{A} + r_1(\cos \phi, \sin \phi). \quad (6)$$

The floating output joint $\mathbf{C}$ must satisfy

$$\|\mathbf{C} - \mathbf{B}\| = r_2, \quad \|\mathbf{C} - \mathbf{D}\| = r_3. \quad (7)$$

It is therefore obtained from the intersection of two circles. The geometry routine explicitly distinguishes separate, contained, tangent, two-solution, coincident, concentric, and negative-radius cases. For two solutions, the solver chooses the candidate nearest the preceding valid $\mathbf{C}$. This local continuity rule preserves the assembly branch during a servo sweep without embedding a fixed “upper” or “lower” branch rule.

The output joint $\mathbf{C}$ is not grounded to the anatomy. Instead, a short structural anchor leaves it at fixed coupler-relative offset δ :

$$\mathbf{E} = \mathbf{C} + L_a \begin{bmatrix} \cos(\theta_c + \delta) \\ \sin(\theta_c + \delta) \end{bmatrix}, \quad \theta_c = \text{atan2}(C_y - B_y, C_x - B_x). \quad (8)$$

Two revolute rectangular drivers continue from $\mathbf{E}$: one carries the middle-phalanx contactor, and the second has an ungrounded right endpoint and carries the distal contactor. This split accommodates two dorsal contacts without introducing an anatomical distal ground attachment.

When no finite intersection exists, only the crank pose is updated. The previous valid coupler and rocker poses remain available for rendering and the state is marked invalid. No non-finite values enter the render loop.

5 Simultaneous finger and contactor solve

The index finger is a serial proximal–middle–distal chain. DIP flexion is coupled to PIP flexion by $\gamma = k\beta$, with $k = 0.62$. For contactor j , let $\mathbf{F}_j(\alpha, \beta)$ be its point on the finger centreline and $\mathbf{n}_{d,j}$ the segment normal directed toward positive world Y . Let W_j be the total internal dorsal-to-flexor ring width. Its admissible domain and per-side affordance are

$$W_j \geq w_{f,j}, \quad a_j = \frac{W_j - w_{f,j}}{2} \geq 0. \quad (9)$$

At the lower bound the ring matches the rendered digit width and provides no offset affordance. The required linkage centreline point is

$$\mathbf{T}_j = \mathbf{F}_j + \left(\frac{W_j}{2} + \frac{w_{l,j}}{2} \right) \mathbf{n}_{d,j}, \quad (10)$$

where $w_{l,j}$ is driver-link width. The unknown vector $\mathbf{z} = (\alpha, \beta, \eta_m, \eta_d)$ contains MCP and PIP angles plus the two passive driver angles. With $\mathbf{P}_m(\mathbf{z})$ and $\mathbf{P}_d(\mathbf{z})$ denoting the two linkage contact points, the default solve drives the four residuals

$$\mathbf{r}(\mathbf{z}) = \begin{bmatrix} \mathbf{P}_m - \mathbf{T}_m \\ \mathbf{P}_d - \mathbf{T}_d \end{bmatrix} = \mathbf{0}. \quad (11)$$

A projected, damped Gauss–Newton iteration uses a finite-difference Jacobian and the previous valid pose as its initial state. When a hand or distal-driver joint reaches a configured limit, that variable remains on its active bound while the remaining proximal and distal variables continue minimizing the contact residual. Thus one blocked joint does not freeze the whole assembly. The tuned default linkage uses a servo range of 38° through 180° ; fine-step tests exercise the entire interval for D2–D5 and reach full configured MCP extension, approximately 85° MCP flexion, and more than 95° PIP flexion. This is a positional IK; no artificial stiffness is required. Rendering places a rectangular pad dorsal to each digit segment and a U-shaped band down to the flexor side.

Every rectangular link is also tested against each phalanx capsule, except when the link carries the contactor explicitly assigned to that same phalanx. That pair is treated as unilateral actuator contact: attempted overlap loads and poses the digit rather than invalidating the whole assembly. For every other link–segment pair, link centreline segment S_l , finger centreline segment S_f , link width w_l , finger width w_f , and clearance ϵ , a pose is admissible only if

$$\text{dist}(S_l, S_f) \geq \frac{w_l + w_f}{2} + \epsilon. \quad (12)$$

The contact solve includes each link half-width in its dorsal offset, so the intended bars remain outside the finger capsules. Links are also tested against the finite dorsal ground-plane segment and base rail. Within the rail’s longitudinal span, a signed half-plane test prevents a servo-driven link from passing to the hand-side of the rail even when it does not intersect the rail. Intended servo and rocker pivot coincidences are permitted; other rail crossings or capsule overlaps are rejected. Rejection restores the last valid link, hand, mount, and contact configuration.

6 Gravity-only static moments

Each phalanx is approximated as a uniform cylinder, consistent with published index-finger and hand parameterizations [4, 6]. For segment i , rendered length L_i , diameter d_i , and tissue density $\rho = 1160 \text{ kg m}^{-3}$, the estimated mass is

$$m_i = \rho \pi \left(\frac{d_i}{2} \right)^2 L_i. \quad (13)$$

The centre of mass is the segment midpoint. With gravity $\mathbf{g} = (0, -g)$ and joint position $\mathbf{J}_k$, the world- z gravity moment from all segments distal to joint k is

$$\tau_k^{g,z} = \sum_{i \in \mathcal{D}(k)} (\mathbf{r}_{i,k} \times m_i \mathbf{g})_z, \quad \mathbf{r}_{i,k} = \mathbf{C}_i - \mathbf{J}_k. \quad (14)$$

The implementation converts this to the stored generalized-angle sign convention. An optional future passive term has the data form

$$\tau_k^s = -K_k(q_k - q_{0,k}), \quad (15)$$

with defaults $K_k = 0$. The posture-holding moment reported by the simulator is

$$\tau_k^{hold} = -(\tau_k^g + \tau_k^s). \quad (16)$$

This is the generalized moment required to maintain the selected posture under gravity only. It is not a tendon-force, contact-force, or joint-reaction-force solution; those require additional muscle and physiological constraints, and are posture dependent [4]. A future configurable stiffness model can populate K_k and $q_{0,k}$ without changing the load decomposition; finger models with torsional joint terms provide a relevant extension path [5].

7 Inverse endpoint controls

Each rectangular link exposes its positive-local- x endpoint as a control handle. Dragging the crank handle directly sets and clamps the servo angle. For constrained four-bar, anchor, middle-driver, and distal-driver handles, the prototype samples the servo ROM and selects the valid angle whose endpoint is nearest the pointer target. This bounded one-dimensional inverse solve preserves the same forward solver as the source of truth. An unconstrained imported link instead rotates about its left endpoint.

User-added links are never floating: creation requires an underlying link or servo context and atomically creates a revolute joint at the new rectangle’s left endpoint. Passive parent–child attachments are propagated after the main four-bar solve so their joint locations follow moving parent links. The optional construction layer renders the full radius- L locus centred on the live left joint of every nonfixed rectangular link, including the anchor, middle, and distal drivers. These centres update after parent constraints, so the displayed circles and the solved joint closures share the same poses.

8 Versioned state export

The JSON exporter serializes a schema name and version, an export timestamp, explicit units, the shared overall hand scale, the active tab, and complete D2 through D5 states. Each digit entry includes its mechanism and hand configuration, independent servo pose, contactor geometry, active joint-limit states, and gravity-moment results. The export is a detached snapshot, so subsequent animation does not mutate already generated data. Angles remain in radians and lengths remain in millimetres.

9 Update ordering and invariants

Simulation time advances independently from drawing. The servo sweeps between its configured limits, reflecting direction at each limit, before the analytic mechanism and finger solves run. Rendering is read-only with respect to mechanism state. The principal invariants are: link poses remain finite; solved endpoint distances agree with link lengths up to floating-point tolerance; and branch choice is based only on the previous valid configuration.

10 Next mathematical milestone

A general constraint graph should replace the four-bar strategy. A practical next step is a graph traversal that resolves analytically solvable dyads and reports degrees of freedom, disconnected components, singularities, and overconstraints. Numerical residual minimization can then serve as a bounded fallback, with the analytic result and prior frame used as its initial condition.

References

- [1] Kai-Nan An, Edmund Y. Chao, William P. Cooney, and Ronald L. Linscheid. Normative model of human hand for biomechanical analysis. *Journal of Biomechanics*, 12(10):775–788, 1979.
- [2] Bryan Buchholz, Thomas J. Armstrong, and Steven A. Goldstein. Anthropometric data for describing the kinematics of the human hand. *Ergonomics*, 35(3):261–273, 1992.
- [3] Gkionoul Nteli Chatzioglou, Yelda Pinar, and Figen Govsa. Biometric analysis hand parameters in young adults for prosthetic hand and ergonomic product applications. *Anatomy & Cell Biology*, 57(2):172–182, 2024.
- [4] Dan Qiu and Derek G. Kamper. Orthopaedic applications of a validated force-based biomechanical model of the index finger. In *2014 36th Annual International Conference of the IEEE Engineering in Medicine and Biology Society*, pages 4013–4016, 2014.
- [5] Panagiotis Tsakonas, Neil D. Evans, Joseph Hardwicke, and Michael J. Chappell. Parameter estimation of a model describing the human fingers. *Healthcare Technology Letters*, 11(1):1–15, 2024.
- [6] Panagiotis Tsakonas, Neil D. Evans, Joseph Hardwicke, Shervanthi Homer-Vanniasinkam, and Michael J. Chappell. In-vivo hand mass determination and an anthropometric investigation on segment length and radius for prosthetic segment design. *Healthcare Technology Letters*, 13(1):e70078, 2026.